Luana Soares

Postdoctoral Researcher · Uppsala University · she/her

Nov/2026

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Research experience

Nov 2026 – present

Postdoctoral Researcher, Birgitta Sintring Fellow · Plant Ecology and Evolution, Department of Ecology and Genetics, Uppsala University, Sweden

Supervisor: Mario Vallejo-Marín. Project: How new polyploid species form: population genomic insights from *Mimulus peregrinus*. Whole-genome resequencing to study subgenome ancestry, homoeologous exchange and selection after recent allopolyploidization.

2026 – present

Postdoctoral Researcher, parallel position (CAPES AURORA Program) · Department of Botany, Federal University of Rio Grande do Sul (UFRGS), Brazil

Supervisor: Caroline Turchetto. Project: Evolutionary history, genetic diversity and conservation of *Nicotiana* sect. *Alatae* in the Campos Sulinos. DArT-seq bioinformatic pipelines, population genomics, gene flow and species delimitation, ecological niche modelling and climate refugia, and training of graduate students in bioinformatics.

2025 – 2026

Postdoctoral Researcher, Evolutionary Genomics · Institute of Plant Molecular and Cellular Biology (IBMCP, CSIC-UPV), Valencia, Spain

Project: Genomic architecture of rapid diversification in Solanaceae lineages. De novo and reference-guided genome assembly and annotation; structural variant and transposable element characterization; phylogenomics and introgression detection; comparative genomics of gene flow and adaptive evolution; reproducible pipelines in Nextflow.

2026

Visiting researcher · Finnish Museum of Natural History, University of Helsinki, Finland

Research visit to Péter Poczai's group, funded by the ESEB Godfrey Hewitt Mobility Award.

2021 – 2025

PhD researcher · Federal University of Rio Grande do Sul (UFRGS), Brazil

Project: The highland's bloom: the impact of fast divergence in highland *Petunia*. Field expeditions and phenotyping, herbarium taxonomy, DNA extraction, genome assembly, population structure, phylogenomics and species delineation, hybridization/introgression detection, demographic inference and environmental association analysis.

Jun – Dec 2023

Visiting PhD researcher · Universitat Politècnica de València, Spain

RAD-seq genome assembly, evaluation of reference genome quality, and comparison of SNP calling across reference mapping strategies.

2019 – 2021

Master's researcher · Federal University of Rio Grande do Sul (UFRGS), Brazil

Project: Phylogeography, genetic diversity and population structuring of *Petunia altiplana*. Fieldwork, cpDNA and SSR genotyping, phylogeography, population genetics and demographic inference.

2014 – 2018

Undergraduate researcher · Federal University of Goiás (UFG), Brazil

Project: How do new enzymes arise? Molecular evolution of new gene copies in the rhodanese superfamily in Diptera.

Education

2021 – 2025

PhD, Genetics and Molecular Biology · UFRGS, Brazil

Thesis: The highland's bloom: the impact of fast divergence in highland *Petunia*. Advisor: Loreta Brandão de Freitas. Visiting student at Universitat Politècnica de València (6 months), advised by Aureliano Bombarely (CAPES-PrInt scholarship).

2019 – 2021 **MSc, Genetics and Molecular Biology** · UFRGS, Brazil
Thesis: Phylogeography, genetic diversity and population structuring of *Petunia altiplana*. Advisor: Loreta Brandão de Freitas.

2014 – 2018 **BSc, Biology** · Federal University of Goiás (UFG), Brazil
Advisor: Iderval da Silva Sobrinho. Visiting student at Instituto Politécnico de Bragança, Portugal (international mobility).

Grants & awards

2026 **Birgitta Sintring Fellowship** · Uppsala University
Personal stipend of 831,600 SEK plus 405,000 SEK in running costs, for the project "How new polyploid species form: population genomic insights from *Mimulus peregrinus*".

2026 **Postdoctoral Fellowship, CAPES AURORA Program** · CAPES, Brazil · UFRGS

2026 **ESEB Godfrey Hewitt Mobility Award (€2,000)** · European Society for Evolutionary Biology
Completed research visit to Péter Poczai, Finnish Museum of Natural History, University of Helsinki.

2025 **Postdoctoral Fellowship (€30,000)** · National Council for Scientific and Technological Development (CNPq), Brazil

2024 **ESEB Conference Travel Award (€1,132)** · European Society for Evolutionary Biology

2023 **CAPES-PrInt Scholarship (€10,590)** · Doctoral stay at Universitat Politècnica de València

Teaching & mentoring

2023 **Workshop instructor: Genetic analysis of population genomics (30 h)** · UNNE-CONICET, Argentina
Designed the curriculum and taught cleaning, filtering and de novo assembly of RAD-seq data.

2022 – 2024 **Undergraduate mentor** · UFRGS, Brazil
Project: Contribution of interspecific hybridization to the genetic diversity of *Petunia* species.

Courses & workshops attended

Jun 2026 **SLiM Workshop: evolutionary modelling with SLiM** · University of Tartu, Estonia
Five-day workshop on individual-based, genetically explicit evolutionary simulation with SLiM, taught by Ben Haller (Messer Lab, Cornell University). 15–19 June 2026.

Skills

Programming
R, Python, shell scripting; workflow managers Nextflow and Snakemake

Genomics
Genome assembly (long and short reads), structural variant calling, phylogenomics, comparative genomics, gene flow and introgression analysis, environmental association

Field & lab
Population sampling and phenotyping, herbarium specimen preparation, DNA extraction, PCR, SSR genotyping

Languages
Portuguese (native), English (proficient), Spanish (proficient)

Publications

Peer-reviewed articles

1. Couto MAMS, **Soares LS**, Rodrigues DM, Burgos VGC, Bombarely A, Freitas LB, Turchetto C (2026). Genomic evidence of hybridization between two species in *Nicotiana* sect. *Alatae*. *Annals of Botany, mcag083*. https://doi.org/10.1093/aob/mcag083
2. Augsten M∗, **Soares LS***, Turchetto C, Stehmann JR, Freitas LB (2026). Extremely rare and endangered: genetic diversity of *Nicotiana gandarela* (Solanaceae). *Botanical Journal of the Linnean Society (accepted)*.
3. **Soares LS**, Bombarely A, Freitas LB (2025). Geographic isolation, ecological adaptation, and hybridization influenced fast divergence of a Solanaceae plant group from subtropical highland grasslands. *Molecular Phylogenetics and Evolution, 213, 108446*. https://doi.org/10.1016/j.ympev.2025.108446
4. **Soares LS**, Stehmann JR, Freitas LB (2025). The genus *Petunia* (Solanaceae): evolutionary synthesis and taxonomic review. *Plants, 14(10), 1478*. https://doi.org/10.3390/plants14101478
5. **Soares LS**, Bombarely A, Freitas LB (2024). How many species are there? Lineage diversification and hidden speciation in Solanaceae from highland grasslands in southern South America. *Annals of Botany, 134(7), 1291–1305*. https://doi.org/10.1093/aob/mcae144
6. Souza AC∗, **Soares LS***, Backes A, Simon L, Pezzi PH, Turchetto C, Freitas LB (2024). Unravelling the genetic diversity and taxonomic ambiguities of endemic *Petunia* species from subtropical highland grasslands. *Botanical Journal of the Linnean Society, boae016*. https://doi.org/10.1093/botlinnean/boae016
7. **Soares LS**, Freitas LB (2024). The phylogeographic journey of a plant species from lowland to highlands during the Pleistocene. *Scientific Reports, 14(1), 3825*. https://doi.org/10.1038/s41598-024-53414-4
8. Simon L∗, **Soares LS***, Freitas LB (2023). Disentangling the causes of high polymorphism sharing in sympatric *Petunia* species from subtropical highland grasslands: insights from nuclear diversity. *Genetics and Molecular Biology, 46(3 suppl 1)*. https://doi.org/10.1590/1678-4685-gmb-2023-0159
9. **Soares LS**, Fagundes NJR, Freitas LB (2023). Past climate changes and geographical barriers: the evolutionary history of a subtropical highland grassland species of Solanaceae, *Petunia altiplana*. *Botanical Journal of the Linnean Society*. https://doi.org/10.1093/botlinnean/boac051

Preprints & manuscripts in revision

1. **Soares LS**, Gonçalves LT, Guzmán-Rodriguez S, Bombarely A, Freitas LB (2026). Reference genome choice impacts SNP recovery but not evolutionary inference in young species. *bioRxiv (in revision)*. https://doi.org/10.64898/2026.02.04.703758
2. **Soares LS**, Fagundes NJR, Bombarely A, Freitas LB (2026). The speciation continuum in bloom: incomplete lineage sorting, gene flow, and reticulate evolution in rapidly diverging plant lineages. *bioRxiv (in revision)*. https://doi.org/10.64898/2026.03.30.715067
3. Couto MAMS∗, **Soares LS***, Augsten M∗, Schnitzler CK, Bombarely A, Stehmann JR, Turchetto C (2026). Unraveling speciation in *Nicotiana* (Solanaceae): a story of environmental shaping. *In revision*.

Theses

1. **Soares LS** (2025). The highland's bloom: the impact of fast divergence in highland *Petunia*. *PhD thesis, Federal University of Rio Grande do Sul (UFRGS)*.
2. **Soares LS** (2021). Phylogeography, genetic diversity and population structuring of *Petunia altiplana*. *MSc thesis, Federal University of Rio Grande do Sul (UFRGS)*.

Conference presentations

1. **Soares LS** [Poster] (2025). The speciation continuum in bloom: incomplete lineage sorting, gene flow, and reticulate evolution in rapidly diverging plant lineages. *Congress of the European Society for Evolutionary Biology (ESEB 2025), Barcelona, Spain*.
2. **Soares LS** [Poster] (2024). When the reference genome does not matter: recent and fast divergence species. *I Congresso Brasileiro de Biologia Evolutiva, Curitiba, Brazil*.
3. **Soares LS** [Talk · ESEB Conference Travel Award] (2024). Exploring the origins: drivers of diversity and ancestral polymorphism in southern South America's highland grasslands. *3rd Joint Congress on Evolutionary Biology, Montreal, Canada*.
4. **Soares LS** [Poster] (2023). The highland bloom: unraveling the evolutionary secrets of a highland *Petunia*. *Evolution 2023, Albuquerque, NM, USA*.
5. **Soares LS** [Poster] (2020). Could a whole valley stop gene flow in *Petunia altiplana*? *SolSeminars (virtual)*.
6. **Soares LS** [Poster] (2020). River deep, mountain high: do geographic barriers restrain gene flow in *Petunia altiplana*? *I Meeting of Systematics, Biogeography and Evolution (virtual)*.